Universal Calculus

The Kleban Operator as Primitive Generator

A formal branched equalizer calculus for latent-to-realspace reconstruction

This document condenses the branched tri-cylindrical construction into a readable operator calculus. The most fundamental operator is the *Kleban Operator*, written $\mathfrak{K}_\beta$. Realspace is its kernel, defect is its value, metric is obtained from its differential, curvature is the coboundary of its gluing connection, and dynamics are variational laws over its norm.

Kleban Operator Edition

June 8, 2026

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Executive compression

The calculus has one primitive generator:

$$\boxed{\mathfrak{K}_\beta = \delta_A \circ \mathcal{R}^\beta.}$$

It sends branch-selected latent projection data to triadic gluing defect. The exact realspace sector is the zero locus:

$$\boxed{\mathcal{M}_0^\beta = \ker \mathfrak{K}_\beta = \mathrm{Eq}(\mathcal{R}_0^\beta, \mathcal{R}_1^\beta, \mathcal{R}_2^\beta).}$$

The finite-resolution realspace sector is the tolerance locus:

$$\boxed{\mathcal{M}_\varepsilon^\beta = \{\lambda : \|\mathfrak{K}_\beta(\lambda, \theta)\| \le \varepsilon\}.}$$

The normal defect metric is the pullback of a defect contrast through the differential of the Kleban Operator:

$$\boxed{g_N^\beta = (D\mathfrak{K}_\beta)^* \Xi (D\mathfrak{K}_\beta).}$$

The tangential realspace metric, when separately specified, is

$$\boxed{g_T^\beta = (D\pi_\beta)^* h (D\pi_\beta).}$$

The relative curvature is another axis coboundary:

$$\boxed{\Omega_\beta = \delta_A d(d\alpha + \omega).}$$

The dynamics are stationary or locally minimizing histories of Kleban defect plus regularity, boundary, and branch-transition costs:

$$\boxed{\delta\mathscr{S} = 0, \qquad \mathscr{S}[H] = \sum_n \left(\|\mathfrak{K}_{\beta_n}\|^2_{\Xi_n} + \lambda_R \mathscr{R}_n + \lambda_\partial \mathscr{B}_n + \lambda_B \,\mathrm{TV}(\beta_n) \right).}$$

Thus the terminal grammar is

$$\boxed{\text{stable comparison} \to \text{branch projection} \to \mathfrak{K}_\beta \to \ker \mathfrak{K}_\beta \to D\mathfrak{K}_\beta \to g, \Omega, \delta\mathscr{S}.}$$

1 Purpose

This document rewrites the branched tri-cylindrical realspace construction as a formal calculus. The intended object is not a prose model, but an operator system with stable definitions, symmetric notation, and reusable manipulation rules.

The word *universal* is used in the syntactic sense: the calculus is written without privileging the names x, y, z, without making the elliptic branch primitive, and without treating realspace as an assumed background. It is not an empirical universality claim. Empirical interpretation would require additional correspondence maps from the formal variables to observations.

The core reduction is:

realspace is not primitive; realspace is the equalizer of branch-selected projections.

The primitive operator that measures failure of this equalization is named the *Kleban Operator*.

2 Primitive objects

2.1 Axis set

Let

$$A = \mathbb{Z}_3 = \{0, 1, 2\}$$

with cyclic convention $i + 1, i + 2 \pmod 3$. The symbols $0, 1, 2$ may later be read as x, y, z, but the calculus itself uses the symmetric cyclic notation.

Let

$$V = \mathbb{R}^3, \qquad \{e_i\}_{i \in A}$$

be the real coordinate target and its standard basis.

2.2 Latent comparison space

At a chart index $n \in I \subseteq \mathbb{N}$, there is a raw state set S_n and an admissible transformation family T_n.

Definition 2.1 (Stable comparison)**.** A comparison datum is

$$C = (a, b, \delta, \chi),$$

where $a, b \in S_n$, δ is a contrast value, and χ is a context locator. For tolerance $\varepsilon \geq 0$, define

$$\mathsf{Stab}_\varepsilon(S_n, T_n) = \{(a, b, \delta, \chi) : |\delta(a, b; \chi) - \delta(Ta, Tb; T\chi)| \leq \varepsilon \quad \forall T \in T_n\}.$$

Definition 2.2 (Latent comparison space)**.** The latent comparison space is the structured quotient generated by stable comparisons:

$$\Lambda_n = \langle \mathsf{Stab}_\varepsilon(S_n, T_n) \rangle / \sim .$$

It is not assumed to be spatial. Differential notation on Λ_n denotes a smooth approximation to the stable comparison structure.

2.3 One global phase

The calculus has exactly one global phase variable:

$$\theta_n = \theta_0 + \omega n$$

in discrete form, or

$$\theta(t) = \theta_0 + \omega t$$

in differentiable approximation. Each axis sector reads the same θ, but may interpret it through a branch-dependent phase law.

Axiom 2.3 (Single phase carrier)**.** There is one phase carrier

$$\theta \in \mathbb{R} \to S^1.$$

Elliptic and polar branches read θ modulo 2π. Hyperbolic branches read the lift in $\mathbb{R}$.

3 Branch geometry

3.1 Branch set

Let the branch set be

$$\mathcal{B} = \{E, P, H\}.$$

The three branches are:

Branch	Name	Phase domain	Transverse signature
E	elliptic fixed-radius leaf	S^1	$u^2 + v^2$
P	polar parent	S^1	$u^2 + v^2$
H	hyperbolic outward leaf	$\mathbb{R}$	$u^2 - v^2$

Define the branch signature

$$\epsilon_b = \begin{cases} +1, & b \in \{E, P\}, \\ -1, & b = H. \end{cases}$$

Define branch trigonometry

$$C_b(\psi) = \begin{cases} \cos\psi, & b \in \{E, P\}, \\ \cosh\psi, & b = H, \end{cases} \qquad S_b(\psi) = \begin{cases} \sin\psi, & b \in \{E, P\}, \\ \sinh\psi, & b = H. \end{cases}$$

Define the branch phase domain

$$\Theta_b = \begin{cases} S^1, & b \in \{E, P\}, \\ \mathbb{R}, & b = H. \end{cases}$$

3.2 Radius law

For each axis $i \in A$, let $R_i > 0$ be the frozen elliptic radius and let

$$r_i : \Lambda_n \to \mathbb{R}_{\geq 0}$$

be the unfrozen radial scale. Define

$$\rho_{i,b}(\lambda) = \begin{cases} R_i, & b = E, \\ r_i(\lambda), & b \in \{P, H\}. \end{cases}$$

3.3 Transverse insertion

Definition 3.1 (Cyclic insertion)**.** The transverse insertion operator is

$$J_i : \mathbb{R}^2 \to V, \qquad J_i(u, v) = ue_{i+1} + ve_{i+2}.$$

It inserts a two-dimensional transverse coordinate pair into the plane perpendicular to axis i.

3.4 Branch trace

Definition 3.2 (Universal branch trace)**.** For $b \in \mathcal{B}$, define

$$\Gamma_b(\rho, \psi, \sigma) = s_b\rho\big(C_b(\psi), S_b(\psi)\big),$$

where

$$s_b = \begin{cases} 1, & b \in \{E, P\}, \\ \sigma \in \{+1, -1\}, & b = H. \end{cases}$$

Thus σ is active only for hyperbolic sheets.

Proposition 3.3 (Branch quadratic identity)**.** *Let* $q_b(u, v) = u^2 + \epsilon_b v^2$. *Then*

$$q_b(\Gamma_b(\rho, \psi, \sigma)) = \rho^2.$$

Proof. For E and P, use $\cos^2\psi + \sin^2\psi = 1$. For H, use $\cosh^2\psi - \sinh^2\psi = 1$ and $\sigma^2 = 1$. □

4 Projection calculus

For each axis $i \in A$, define latent coordinate fields

$$a_i : \Lambda_n \to \mathbb{R}, \qquad r_i : \Lambda_n \to \mathbb{R}_{\geq 0}, \qquad \alpha_i : \Lambda_n \to \mathbb{R}.$$

The phase read by axis sector i is

$$\psi_i(\lambda, \theta) = \theta + \alpha_i(\lambda).$$

Definition 4.1 (Branch field)**.** A branch field is

$$\beta : \Lambda_n \to \mathcal{B}^A, \qquad \beta(\lambda) = (\beta_0(\lambda), \beta_1(\lambda), \beta_2(\lambda)).$$

Each axis has a selected branch at each latent point.

Definition 4.2 (Axis reconstruction map)**.** The branch-selected reconstruction map along axis i is

$$\boxed{\mathcal{R}_i^\beta(\lambda, \theta) = a_i(\lambda)e_i + J_i\Gamma_{\beta_i(\lambda)}\left(\rho_{i,\beta_i}(\lambda), \theta + \alpha_i(\lambda), \sigma_i(\lambda)\right).}$$

The full reconstruction triple is

$$\mathcal{R}^\beta = (\mathcal{R}_0^\beta, \mathcal{R}_1^\beta, \mathcal{R}_2^\beta) : \Lambda_n \times \Theta \to V^A.$$

This single equation replaces all component-specific x, y, z formulas. Component formulas are given in Appendix A for readability.

5 The Kleban Operator

5.1 Axis coboundary

Definition 5.1 (Axis coboundary)**.** Define

$$\delta_A : V^A \to V^A$$

by

$$(\delta_A R)_i = R_i - R_{i+1}, \qquad i \in A.$$

Equivalently,

$$\delta_A(R_0, R_1, R_2) = (R_0 - R_1, R_1 - R_2, R_2 - R_0).$$

5.2 Primitive operator

Definition 5.2 (Kleban Operator)**.** The Kleban Operator is

$$\boxed{\mathfrak{K}_\beta : \Lambda_n \times \Theta \to V^A, \qquad \mathfrak{K}_\beta(\lambda, \theta) = \delta_A \mathcal{R}^\beta(\lambda, \theta).}$$

Expanded:

$$\boxed{\mathfrak{K}_\beta(\lambda, \theta) = \big(\mathcal{R}_0^\beta - \mathcal{R}_1^\beta, \mathcal{R}_1^\beta - \mathcal{R}_2^\beta, \mathcal{R}_2^\beta - \mathcal{R}_0^\beta\big).}$$

It is the primitive generator of the calculus. It measures the failure of the three branch-selected projection sectors to agree.

Remark 5.3 (Why this is primitive)*.* Every major object is derived from $\mathfrak{K}_\beta$:

$$\text{defect} = \mathfrak{K}_\beta, \qquad \text{realspace} = \ker \mathfrak{K}_\beta, \qquad \text{normal metric} = (D\mathfrak{K}_\beta)^* \Xi (D\mathfrak{K}_\beta), \qquad \text{dynamics} = \operatorname{Crit} \|\mathfrak{K}_\beta\|^2 + \cdots .$$

Definition 5.4 (Full stabilized Kleban Operator)**.** When the stable-comparison origin must be shown explicitly, define

$$\widehat{\mathfrak{K}}_\beta = \mathfrak{K}_\beta \circ \iota_\varepsilon,$$

where

$$\iota_\varepsilon : \langle \mathsf{Stab}_\varepsilon(S_n, T_n) \rangle / \sim \hookrightarrow \Lambda_n.$$

The hatted operator exposes the stable substrate. The unhatted operator is the operational form used after Λ_n has been constructed.

6 Realspace as kernel

Definition 6.1 (Exact realspace locus)**.** The exact branch-realspace locus is

$$\boxed{\mathcal{M}_0^\beta(n) = \ker \mathfrak{K}_\beta = \{\lambda \in \Lambda_n : \mathfrak{K}_\beta(\lambda, \theta_n) = 0\}.}$$

Proposition 6.2 (Equalizer law)**.**

$$\boxed{\ker \mathfrak{K}_\beta = \operatorname{Eq}(\mathcal{R}_0^\beta, \mathcal{R}_1^\beta, \mathcal{R}_2^\beta).}$$

Proof. If $\mathfrak{K}_\beta = 0$, then $\mathcal{R}_0^\beta = \mathcal{R}_1^\beta$, $\mathcal{R}_1^\beta = \mathcal{R}_2^\beta$, and $\mathcal{R}_2^\beta = \mathcal{R}_0^\beta$. Hence all three agree. Conversely, if all three agree, each cyclic difference is zero. ∎

Definition 6.3 (Approximate realspace locus)**.** For tolerance $\varepsilon \geq 0$,

$$\boxed{\mathcal{M}^{\beta}_{\varepsilon}(n) = \{\lambda \in \Lambda_n : \|\mathfrak{K}_{\beta}(\lambda, \theta_n)\| \leq \varepsilon\}.}$$

The exact model is recovered at $\varepsilon = 0$.

Definition 6.4 (Realspace image)**.** On $\mathcal{M}^{\beta}_{0}$, define

$$\pi_{\beta} : \mathcal{M}^{\beta}_{0} \to V, \qquad \pi_{\beta}(\lambda) = \mathcal{R}^{\beta}_{i}(\lambda, \theta_n),$$

where i is arbitrary because all three sectors agree. The effective realspace image is

$$\boxed{U^{\beta}_{\mathrm{real}}(n) = \pi_{\beta}(\mathcal{M}^{\beta}_{0}).}$$

7 Differential calculus of the Kleban Operator

Assume Λ_n is replaced by a differentiable approximation. Let

$$T_{\lambda}\Lambda_n$$

be the tangent space at λ.

Definition 7.1 (Kleban differential)**.** The Kleban differential is

$$\boxed{L_{\beta,\lambda} = D\mathfrak{K}_{\beta}|_{\lambda} : T_{\lambda}\Lambda_n \to T_{\mathfrak{K}_{\beta}(\lambda)}V^{A}.}$$

It is the linearized defect response to infinitesimal latent variation.

Definition 7.2 (Kleban tangent condition)**.** At an exact realspace point $\lambda \in \mathcal{M}^{\beta}_{0}$, an infinitesimal variation $v \in T_{\lambda}\Lambda_n$ is tangent to the exact realspace locus when

$$\boxed{D\mathfrak{K}_{\beta}(v) = 0.}$$

Thus

$$T_{\lambda}\mathcal{M}^{\beta}_{0} = \ker(D\mathfrak{K}_{\beta}|_{\lambda})$$

whenever $\mathcal{M}^{\beta}_{0}$ is a regular submanifold.

Definition 7.3 (Defect-normal direction)**.** A tangent vector $v \in T_{\lambda}\Lambda_n$ is defect-normal relative to $\mathfrak{K}_{\beta}$ when

$$D\mathfrak{K}_{\beta}(v) \neq 0.$$

It exits compatibility to first order.

8 Metric calculus

8.1 Defect contrast

Let Ξ_{λ} be a symmetric bilinear contrast form on the defect target V^{A}:

$$\Xi_{\lambda} : T_{\mathfrak{K}_{\beta}(\lambda)}V^{A} \times T_{\mathfrak{K}_{\beta}(\lambda)}V^{A} \to \mathbb{R}.$$

The symbol Ξ is used instead of K so that the letter K remains reserved for the Kleban Operator.

8.2 Normal metric

Definition 8.1 (Kleban normal metric)**.** The defect-normal metric is

$$\boxed{g_N^\beta = (D\mathfrak{K}_\beta)^*\Xi(D\mathfrak{K}_\beta).}$$

Equivalently,

$$g_{N,\lambda}^\beta(u,v) = \Xi_\lambda(D\mathfrak{K}_\beta(u), D\mathfrak{K}_\beta(v)).$$

Remark 8.2 (Interpretation). g_N^β measures stiffness against leaving compatibility. On the exact kernel, every exact tangent direction satisfies $D\mathfrak{K}_\beta(v) = 0$, so $g_N^\beta(v,v) = 0$ for those directions. Therefore g_N^β is a normal defect metric, not automatically the intrinsic metric of realspace.

8.3 Tangential realspace metric

Definition 8.3 (Tangential metric)**.** Let h be a target contrast on V. The tangential realspace metric is

$$\boxed{g_T^\beta = (D\pi_\beta)^* h(D\pi_\beta).}$$

Equivalently,

$$g_{T,\lambda}^\beta(u,v) = h(D\pi_\beta(u), D\pi_\beta(v)).$$

Definition 8.4 (Metric admissibility)**.** A tangent sector $E_\lambda \subseteq T_\lambda\Lambda_n$ is metrically admissible when

$$\ker(D\mathfrak{K}_\beta|_{E_\lambda}) = 0 \quad \text{modulo gauge directions}$$

for normal metrics, or when

$$\ker(D\pi_\beta|_{E_\lambda}) = 0 \quad \text{modulo gauge directions}$$

for tangential metrics.

9 Connection and curvature calculus

9.1 Gluing connection

Let

$$\alpha = (\alpha_i)_{i\in A}$$

be local phase offsets. Let

$$\omega = (\omega_i)_{i\in A}$$

be additional local transport contributions.

Definition 9.1 (Axis connection)**.** The axis gluing one-forms are

$$\boxed{\mathcal{A}_i = d\alpha_i + \omega_i.}$$

The connection triple is

$$\mathcal{A} = (\mathcal{A}_0, \mathcal{A}_1, \mathcal{A}_2).$$

Definition 9.2 (Axis curvature)**.** The axis curvature two-forms are

$$\boxed{\mathcal{F}_i = d\mathcal{A}_i = d(d\alpha_i + \omega_i).}$$

Because $d^2 = 0$, this reduces to

$$\mathcal{F}_i = d\omega_i$$

whenever α_i is globally smooth.

9.2 Relative curvature

Definition 9.3 (Relative Kleban curvature)**.** The relative curvature is

$$\boxed{\Omega_\beta = \delta_A \mathcal{F}.}$$

Expanded:

$$\Omega_\beta = (\mathcal{F}_0 - \mathcal{F}_1, \mathcal{F}_1 - \mathcal{F}_2, \mathcal{F}_2 - \mathcal{F}_0).$$

Definition 9.4 (Flat relative gluing)**.** The branch-selected gluing is relatively flat when

$$\boxed{\Omega_\beta = 0.}$$

This means all three axis sectors have identical local curvature up to admissible gauge.

10 Gauge laws

10.1 Global phase gauge

The physical phase read by sector i is

$$\psi_i = \theta + \alpha_i.$$

Thus the transformation

$$\boxed{\theta \mapsto \theta + c, \qquad \alpha_i \mapsto \alpha_i - c}$$

leaves every ψ_i invariant.

10.2 Local connection gauge

For local functions $\eta_i : \Lambda_n \to \mathbb{R}$,

$$\boxed{\alpha_i \mapsto \alpha_i + \eta_i, \qquad \omega_i \mapsto \omega_i - d\eta_i}$$

leaves

$$\mathcal{A}_i = d\alpha_i + \omega_i$$

invariant.

Proposition 10.1 (Gauge invariance of Kleban data)**.** *If the phase reads $\psi_i = \theta + \alpha_i$ and branch choices are unchanged, then the global phase gauge leaves $\mathcal{R}_i^\beta$, $\mathfrak{K}_\beta$, $\mathcal{M}_0^\beta$, and U_{real}^β invariant.*

Proof. The substitution sends ψ_i to $(\theta + c) + (\alpha_i - c) = \theta + \alpha_i$. Therefore every branch trace and reconstruction map is unchanged. All derived objects are unchanged. □

11 Branch admissibility calculus

Definition 11.1 (Branch variation)**.** Let $\beta : \Lambda_n \to \mathcal{B}^A$ be discrete. Its branch variation is represented by an interface measure

$$\mathrm{TV}(\beta)$$

in the smooth approximation, or by a weighted count of branch changes across neighboring comparison cells in the discrete approximation.

Axiom 11.2 (Branch stability)**.** A branch field is admissible only when it is locally stable on open compatible regions and changes only across persistent defect interfaces, prescribed boundaries, or prescribed singular strata:

$$\boxed{\operatorname{supp}(d\beta) \subseteq \{\lambda : \|\mathfrak{K}_\beta(\lambda,\theta)\| > \eta\} \cup \partial\Lambda_n \cup \Sigma_{\text{prescribed}}.}$$

Definition 11.3 (Branch transition cost)**.** The branch-transition cost is

$$J_\beta = \lambda_B \operatorname{TV}(\beta)$$

for some weight $\lambda_B \geq 0$, or a more refined interface functional when topology, curvature, or sheet changes require distinct penalties.

Remark 11.4 (No flicker principle). Without branch admissibility, the system could reduce defect by rapidly changing branches rather than by producing coherent gluing. Branch variation is therefore a structural cost, not an optional aesthetic term.

12 Variational calculus

12.1 State and history

A chart state is

$$X_n = (S_n, \Lambda_n, \theta_n, \beta_n, \mathcal{R}^{\beta_n}, \mathcal{A}_n, \mathcal{F}_n, \mathfrak{K}_{\beta_n}, \Xi_n).$$

A history is

$$H = (X_n)_{n\in I}.$$

12.2 Action

Definition 12.1 (Universal Kleban action)**.** The action is

$$\boxed{\mathscr{S}[H] = \sum_{n\in I} \left[\|\mathfrak{K}_{\beta_n}(\lambda_n,\theta_n)\|^2_{\Xi_n} + \lambda_R \mathscr{R}_n + \lambda_\partial \mathscr{B}_n + \lambda_B \operatorname{TV}(\beta_n)\right].}$$

Here:

- $\|\mathfrak{K}_{\beta_n}\|^2_{\Xi_n}$ is gluing defect cost;
- $\mathscr{R}_n$ is regularity cost;
- $\mathscr{B}_n$ is boundary cost;
- $\operatorname{TV}(\beta_n)$ is branch transition cost;
- $\lambda_R, \lambda_\partial, \lambda_B \geq 0$ are weights.

Definition 12.2 (Valid history)**.** A valid history is stationary or locally minimizing:

$$\boxed{H^\star \in \operatorname{Crit}(\mathscr{S}), \qquad \delta\mathscr{S}[H^\star] = 0.}$$

Definition 12.3 (Discrete update)**.** A discrete local update is

$$\boxed{X_{n+1} \in \arg\min_{X'\in\operatorname{Adm}(X_n)} \left[\|\mathfrak{K}_{\beta'}(X')\|^2_{\Xi_n} + \lambda_R \mathscr{R}(X') + \lambda_\partial \mathscr{B}(X') + \lambda_B \operatorname{TV}(\beta')\right].}$$

12.3 Euler operator

When a differentiable variational approximation exists, define the Kleban Euler operator

$$\boxed{\mathcal{E}_{\mathfrak{K}}[H] = \frac{\delta \mathscr{S}}{\delta H}.}$$

The field equation is

$$\boxed{\mathcal{E}_{\mathfrak{K}}[H] = 0.}$$

13 Symmetry calculus

13.1 Axis permutation symmetry

Let S_3 act on $A = \{0, 1, 2\}$ by permuting axis labels. For $\pi \in S_3$,

$$(\pi R)_i = R_{\pi^{-1}(i)}, \qquad (\pi \beta)_i = \beta_{\pi^{-1}(i)}.$$

Proposition 13.1 (Equivariance)**.** *If the insertion maps, contrast forms, and penalties are transformed coherently under $\pi \in S_3$, then*

$$\boxed{\mathfrak{K}_{\pi\beta}(\pi\lambda, \theta) = \pi\mathfrak{K}_{\beta}(\lambda, \theta).}$$

Consequently,

$$\boxed{\mathcal{M}_0^{\pi\beta} = \pi\mathcal{M}_0^{\beta}.}$$

13.2 Universal symmetry group

The natural symmetry group of the calculus is

$$\boxed{G_{\mathrm{UC}} = S_3 \ltimes G_\theta \ltimes G_{\mathcal{A}}.}$$

Here:

- S_3 permutes axis sectors;
- G_θ shifts the global phase with compensating offset shifts;
- $G_{\mathcal{A}}$ is the local connection gauge preserving $\mathcal{A}_i = d\alpha_i + \omega_i$.

Definition 13.2 (Symmetric action)**.** The action is axis-symmetric when

$$\boxed{\mathscr{S}[\pi H] = \mathscr{S}[H] \quad \forall \pi \in S_3.}$$

This requires symmetric defect contrast, regularity, boundary, and branch-transition penalties.

14 Core calculus laws

Rule 14.1 (Kleban kernel law)**.**

$$\mathfrak{K}_\beta = 0 \quad \Longleftrightarrow \quad \mathcal{R}_0^\beta = \mathcal{R}_1^\beta = \mathcal{R}_2^\beta.$$

Rule 14.2 (Tolerance law)**.**

$$\lambda \in \mathcal{M}_\varepsilon^\beta \quad \Longleftrightarrow \quad \|\mathfrak{K}_\beta(\lambda, \theta)\| \le \varepsilon.$$

Rule 14.3 (Linearization law).

$$T_\lambda \mathcal{M}_0^\beta = \ker(D\mathfrak{K}_\beta|_\lambda)$$

whenever $\mathcal{M}_0^\beta$ is regular.

Rule 14.4 (Normal metric law).

$$g_N^\beta(u, v) = \Xi(D\mathfrak{K}_\beta(u), D\mathfrak{K}_\beta(v)).$$

Rule 14.5 (Tangential metric law).

$$g_T^\beta(u, v) = h(D\pi_\beta(u), D\pi_\beta(v)).$$

Rule 14.6 (Curvature coboundary law).

$$\Omega_\beta = \delta_A d(d\alpha + \omega).$$

Rule 14.7 (Gauge law). The transformations

$$\theta \mapsto \theta + c, \quad \alpha_i \mapsto \alpha_i - c$$

and

$$\alpha_i \mapsto \alpha_i + \eta_i, \quad \omega_i \mapsto \omega_i - d\eta_i$$

preserve the corresponding phase reads and gluing connection.

Rule 14.8 (Branch admissibility law).

$$\operatorname{supp}(d\beta) \subseteq \{\|\mathfrak{K}_\beta\| > \eta\} \cup \partial\Lambda \cup \Sigma_{\text{prescribed}}.$$

Rule 14.9 (Stationarity law).

$$\delta\mathscr{S} = 0.$$

15 Operator dictionary

Object	Symbol	Meaning
Stable comparison selector	$\mathsf{Stab}_\varepsilon$	Keeps only contrast data invariant under admissible transformations up to tolerance.
Latent comparison space	Λ	Structured domain generated by stable comparisons; not assumed spatial.
Axis set	$A = \mathbb{Z}_3$	Symmetric three-sector index set.
Branch set	$\mathcal{B} = \{E, P, H\}$	Elliptic fixed leaf, polar parent, hyperbolic outward leaf.
Cyclic insertion	J_i	Inserts a transverse pair into the plane perpendicular to axis i.
Branch trace	Γ_b	Unified circular/hyperbolic transverse parametrization.
Axis reconstruction	$\mathcal{R}_i^\beta$	Branch-selected map from latent data to target space.

Object	Symbol	Meaning
Axis coboundary	δ_A	Takes cyclic projection differences.
Kleban Operator	$\mathfrak{K}_\beta$	Primitive defect generator $\delta_A \mathcal{R}^\beta$.
Exact realspace	$\mathcal{M}_0^\beta$	Kernel of the Kleban Operator.
Approximate realspace	$\mathcal{M}_\varepsilon^\beta$	Tolerance locus of the Kleban Operator.
Realspace image	U_{real}^β	Common image of the agreeing projection sectors.
Kleban differential	$D\mathfrak{K}_\beta$	Linearized defect response.
Defect contrast	Ξ	Bilinear form on defect variations.
Normal metric	g_N^β	Pullback of Ξ through $D\mathfrak{K}_\beta$.
Tangential metric	g_T^β	Pullback of target metric h through $D\pi_\beta$.
Connection	$\mathcal{A}_i$	Local gluing one-form $d\alpha_i + \omega_i$.
Curvature	$\mathcal{F}_i$	Axis curvature $d\mathcal{A}_i$.
Relative curvature	Ω_β	Axis coboundary $\delta_A \mathcal{F}$.
Branch variation	$\mathrm{TV}(\beta)$	Interface measure or discrete branch-change cost.
Action	$\mathscr{S}$	Variational defect-plus-regularity functional.
Euler operator	$\mathcal{E}_{\mathfrak{K}}$	Variational derivative $\delta\mathscr{S}/\delta H$.

16 Construction algorithm

Construction 16.1 (Build a local model)**.** A local model of Universal Calculus is built by the following steps.

1. Choose an index set $I \subseteq \mathbb{N}$.
2. For each $n \in I$, choose a raw state set S_n.
3. Choose admissible transformations T_n.
4. Define stable comparisons $\mathsf{Stab}_\varepsilon(S_n, T_n)$.
5. Generate the latent comparison space Λ_n.
6. Choose the one global phase θ_n.
7. Choose the branch field $\beta : \Lambda_n \to \mathcal{B}^A$.
8. Choose coordinate fields a_i, r_i, α_i and sheet fields σ_i.
9. Construct $\mathcal{R}_i^\beta$ from the universal branch trace.
10. Apply the Kleban Operator $\mathfrak{K}_\beta = \delta_A \mathcal{R}^\beta$.
11. Define realspace as $\mathcal{M}_0^\beta = \ker \mathfrak{K}_\beta$ or $\mathcal{M}_\varepsilon^\beta = \{\|\mathfrak{K}_\beta\| \le \varepsilon\}$.
12. Define the realspace image $U_{\text{real}}^\beta = \pi_\beta(\mathcal{M}_0^\beta)$.
13. Define gluing connection $\mathcal{A}_i = d\alpha_i + \omega_i$.

14. Define relative curvature $\Omega_\beta = \delta_A d\mathcal{A}$.
15. Define defect contrast Ξ and metrics g_N^β, g_T^β as needed.
16. Define branch admissibility through $\mathrm{TV}(\beta)$.
17. Define the action $\mathscr{S}$.
18. Select histories satisfying $\delta\mathscr{S} = 0$ or the discrete local minimization rule.

17 Minimal tuple

Definition 17.1 (Universal Calculus model)**.** A model of Universal Calculus is the tuple

$$\boxed{\mathcal{U}_{\mathfrak{K}} = (I, S, T, \mathsf{Stab}_\varepsilon, \Lambda, \theta, A, \mathcal{B}, \beta, J, \Gamma, \mathcal{R}^\beta, \delta_A, \mathfrak{K}_\beta, \mathcal{M}_0^\beta, \mathcal{M}_\varepsilon^\beta, \pi_\beta, \mathcal{A}, \Omega_\beta, \Xi, g_N^\beta, g_T^\beta, \mathscr{S}).}$$

A local universe is a history of such models satisfying branch admissibility and stationarity.

The shortest complete declaration is:

$$\boxed{\begin{aligned} \mathfrak{K}_\beta &= \delta_A \mathcal{R}^\beta, \\ \mathcal{R}_i^\beta &= a_i e_i + J_i \Gamma_{\beta_i}(\rho_{i,\beta_i}, \theta + \alpha_i, \sigma_i), \\ \mathcal{M}_0^\beta &= \ker \mathfrak{K}_\beta, \\ g_N^\beta &= (D\mathfrak{K}_\beta)^* \Xi (D\mathfrak{K}_\beta), \\ \Omega_\beta &= \delta_A d(d\alpha + \omega), \\ \delta\mathscr{S} &= 0. \end{aligned}}$$

18 Fundamental theorem of the calculus

Theorem 18.1 (Kleban reconstruction theorem)**.** *Given a stable latent comparison space Λ_n, one global phase θ_n, a branch field β, and reconstruction maps $\mathcal{R}_i^\beta$ constructed from $J_i\Gamma_{\beta_i}$, the Kleban Operator*

$$\mathfrak{K}_\beta = \delta_A \mathcal{R}^\beta$$

canonically determines:

1. *the exact realspace locus $\mathcal{M}_0^\beta = \ker \mathfrak{K}_\beta$;*
2. *the approximate realspace locus $\mathcal{M}_\varepsilon^\beta = \{\|\mathfrak{K}_\beta\| \le \varepsilon\}$;*
3. *the sector-independent realspace projection π_β on $\mathcal{M}_0^\beta$;*
4. *the defect-normal metric $g_N^\beta = (D\mathfrak{K}_\beta)^* \Xi (D\mathfrak{K}_\beta)$ after a defect contrast Ξ is supplied;*
5. *the tangent condition $T_\lambda \mathcal{M}_0^\beta = \ker(D\mathfrak{K}_\beta|_\lambda)$ at regular exact points;*
6. *the defect term of the action $\|\mathfrak{K}_\beta\|_\Xi^2$.*

Proof. Items 1 and 2 are definitions of the zero and tolerance loci of $\mathfrak{K}_\beta$. Item 3 follows from the equalizer law. Item 4 follows from pullback of the supplied contrast through the Kleban differential. Item 5 is the standard tangent-kernel condition for a regular level set. Item 6 is the quadratic defect cost generated by the same primitive operator. □

19 Known unknowns and unknown unknowns

19.1 Known unknowns

The calculus exposes the following unresolved formal choices.

1. **Nondegeneracy classification.** The null space of $D\mathfrak{K}_\beta$ must be separated into gauge, physical tangent, and inadmissible degeneracy.
2. **Equalizer existence.** $\ker \mathfrak{K}_\beta$ may be empty, singular, disconnected, or branch-dependent.
3. **Branch penalty non-uniqueness.** Different choices of $\mathrm{TV}(\beta)$ may produce different stable defect interfaces.
4. **Metric signature.** Hyperbolic branches and indefinite defect contrasts may produce pseudo-Riemannian signatures. A signature selection rule is not contained in the bare calculus.
5. **Observable correspondence.** The calculus constructs effective realspace. It does not by itself identify measurement variables.
6. **Finite tolerance behavior.** The relation between $\mathcal{M}_\varepsilon^\beta$ and $\mathcal{M}_0^\beta$ may fail to be stable under branch transitions or singular limits.

19.2 Unknown unknowns

The following failure modes are not ruled out by the formal definitions.

1. branch caustics where β changes faster than the differentiable approximation can resolve;
2. monodromy from transporting around persistent defect interfaces;
3. hidden nonlocality induced by the single global phase carrier;
4. chaotic branch minimizers in the discrete update rule;
5. topology change of $\ker \mathfrak{K}_\beta$ under small changes in Ξ, $\mathrm{TV}(\beta)$, or boundary terms;
6. singular sheets in the hyperbolic branch where admissibility of σ changes.

20 Appendix A: Component formulas

This appendix rewrites the symmetric formula in the familiar x, y, z axis notation. Let $A = \{x, y, z\}$ and use cyclic transverse planes.

For the x-axis:

$$\mathcal{R}_x^E = (a_x, R_x \cos\psi_x, R_x \sin\psi_x),$$

$$\mathcal{R}_x^P = (a_x, r_x \cos\psi_x, r_x \sin\psi_x),$$

$$\mathcal{R}_x^H = (a_x, \sigma_x r_x \cosh\psi_x, \sigma_x r_x \sinh\psi_x).$$

For the y-axis:

$$\mathcal{R}_y^E = (R_y \sin\psi_y, a_y, R_y \cos\psi_y),$$

$$\mathcal{R}_y^P = (r_y \sin\psi_y, a_y, r_y \cos\psi_y),$$

$$\mathcal{R}_y^H = (\sigma_y r_y \sinh\psi_y, a_y, \sigma_y r_y \cosh\psi_y).$$

For the z-axis:

$$\mathcal{R}_z^E = (R_z \cos\psi_z, R_z \sin\psi_z, a_z),$$

$$\mathcal{R}_z^P = (r_z \cos\psi_z, r_z \sin\psi_z, a_z),$$
$$\mathcal{R}_z^H = (\sigma_z r_z \cosh\psi_z, \sigma_z r_z \sinh\psi_z, a_z).$$

The Kleban Operator is then

$$\mathfrak{K}_\beta = (\mathcal{R}_x^\beta - \mathcal{R}_y^\beta, \mathcal{R}_y^\beta - \mathcal{R}_z^\beta, \mathcal{R}_z^\beta - \mathcal{R}_x^\beta).$$

21 Appendix B: One-page specification

Objects

$$A = \mathbb{Z}_3, \quad V = \mathbb{R}^3, \quad \mathcal{B} = \{E, P, H\}, \quad \Lambda = \langle \mathsf{Stab}_\varepsilon(S,T)\rangle / \sim, \quad \theta \in \mathbb{R} \to S^1.$$

Branch trace

$$\Gamma_b(\rho,\psi,\sigma) = s_b\rho(C_b\psi, S_b\psi), \quad q_b(\Gamma_b) = \rho^2.$$

Projection

$$\mathcal{R}_i^\beta = a_i e_i + J_i \Gamma_{\beta_i}(\rho_{i,\beta_i}, \theta + \alpha_i, \sigma_i).$$

Primitive operator

$$\boxed{\mathfrak{K}_\beta = \delta_A \mathcal{R}^\beta.}$$

Realspace

$$\mathcal{M}_0^\beta = \ker \mathfrak{K}_\beta, \qquad \mathcal{M}_\varepsilon^\beta = \{\|\mathfrak{K}_\beta\| \le \varepsilon\}, \qquad U_{\mathrm{real}}^\beta = \pi_\beta(\mathcal{M}_0^\beta).$$

Metric

$$g_N^\beta = (D\mathfrak{K}_\beta)^* \Xi (D\mathfrak{K}_\beta), \qquad g_T^\beta = (D\pi_\beta)^* h (D\pi_\beta).$$

Curvature

$$\mathcal{A}_i = d\alpha_i + \omega_i, \qquad \mathcal{F}_i = d\mathcal{A}_i, \qquad \Omega_\beta = \delta_A \mathcal{F}.$$

Dynamics

$$\mathscr{S} = \sum_n \left(\|\mathfrak{K}_{\beta_n}\|_{\Xi_n}^2 + \lambda_R \mathscr{R}_n + \lambda_\partial \mathscr{B}_n + \lambda_B \,\mathrm{TV}(\beta_n) \right), \qquad \delta\mathscr{S} = 0.$$

Source note

This formal calculus edition condenses the branched tri-cylindrical construction of local realspace into a symmetric operator system. It preserves the central ingredients: stable comparisons, one global phase, elliptic/polar/hyperbolic branches, equalizer-defined realspace, gluing defect, curvature, pullback metric, branch admissibility, and variational dynamics.